# YEAR 12 GENERAL MATHEMATICS INVESTIGATIVE TASK 4

## Statistics

**Question 1**

Data obtained from a group of people at a local health club are shown in the following table.

| | Sex | Height (cm) | Weight (kg) | Fat (%) | Smoker |
|---|---|---|---|---|---|
| 1 | M | 184 | 75.2 | 15.6 | N |
| 2 | F | 161 | 60.3 | 29.6 | N |
| 3 | F | 172 | 66.2 | 26.2 | Y |
| 4 | M | 189 | 92.6 | 27.2 | N |
| 5 | M | 179 | 76.6 | 19.2 | N |
| 6 | M | 172 | 83.5 | 8.2 | N |
| 7 | M | 184 | 68.9 | 9.1 | N |
| 8 | F | 165 | 63.7 | 22.3 | N |
| 9 | F | 169 | 62.4 | 24.6 | N |
| 10 | F | 158 | 51.1 | 18.3 | Y |
| 11 | M | 192 | 88.7 | 23.7 | N |
| 12 | F | 165 | 68.4 | 31.3 | N |
| 13 | M | 176 | 82.0 | 26.3 | Y |
| 14 | F | 170 | 55.2 | 20.8 | N |
| 15 | F | 161 | 59.8 | 25.1 | N |
| 16 | M | 182 | 73.5 | 19.2 | N |
| 17 | F | 167 | 53.9 | 19.9 | Y |
| 18 | M | 170 | 59.8 | 15.2 | N |
| 19 | M | 180 | 81.3 | 26.2 | N |
| 20 | M | 183 | 77.6 | 21.5 | N |
| 21 | F | 162 | 58.1 | 24.7 | N |
| 22 | F | 167 | 62.3 | 25.2 | N |
| 23 | F | 170 | 60.8 | 23.1 | Y |
| 24 | M | 186 | 89.5 | 11.6 | N |
| 25 | F | 159 | 50.7 | 20.6 | N |
| 26 | F | 164 | 59.7 | 24.7 | N |
| 27 | M | 177 | 68.9 | 18.2 | Y |
| 28 | F | 186 | 72.1 | 23.2 | N |
| 29 | F | 167 | 62.2 | 26.9 | N |
| 30 | F | 165 | 64.8 | 23.5 | Y |
| 31 | M | 187 | 86.2 | 12.6 | N |
| 32 | F | 159 | 63.2 | 30.6 | N |
| 33 | M | 169 | 73.2 | 11.8 | N |

*Continued*

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| | Sex | Height (cm) | Weight (kg) | Fat (%) | Smoker |
|---|---|---|---|---|---|
| 34 | F | 163 | 54.0 | 21.7 | N |
| 35 | M | 185 | 82.3 | 22.9 | N |
| 36 | M | 167 | 76.4 | 27.1 | N |
| 37 | M | 184 | 77.9 | 16.2 | N |
| 38 | F | 168 | 62.5 | 28.3 | N |
| 39 | F | 163 | 57.8 | 27.1 | N |
| 40 | M | 186 | 89.6 | 26.9 | N |

Consider the heights of the women at the health club.

**a** What type of data is this?

**1 mark**

**b** Find the mean, median and modal height, correct to one decimal place.

**3 marks**

**c** Find the range and the standard deviation of the heights, correct to one decimal place.

**2 marks**

**d** To display the data, create a frequency table with class intervals of 5 cm.

**2 marks**

**e** Draw a histogram to represent the data.

**2 marks**

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**f** Describe the shape of the histogram.

**1 mark**

**g** Comment on the distribution of the heights of the women at this health club.

**2 marks**

**Total 13 marks**

**Question 2**

Now, consider the heights of the men at the health club.

**a** Find the mean, median and modal height, correct to one decimal place.

**3 marks**

**b** Find the range and the standard deviation of the heights, correct to one decimal place.

**2 marks**

**c** Create a stem-and-leaf plot to represent the data.

**2 marks**

**d** Draw a pair of parallel boxplots to represent the two sets of data (both men and women), demonstrating and marking in the presence of any outliers.

**4 marks**

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**e** Use what you have found to compare the two data sets.

**2 marks**

**Total 13 marks**

**Question 3**

The owner of the health club is about to introduce a new program aimed at helping people to quit smoking, but first needs to assess the smoking status of its members. The owner wants to determine whether there is a relationship between gender and smoking status.

**a** Identify the explanatory and response variables in this potential relationship.

**1 mark**

**b** Create a two-way table comparing gender and smoking status.

**2 marks**

**c** What percentage of:

**i** smokers are male

**ii** females are smokers

**iii** the group are smokers?

**3 marks**

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**d** Convert the two-way table into a percentage two-way table and hence draw a segmented column graph showing the smoking habits of each gender.

**3 marks**

**e** Does there appear to be a relationship between gender and smoking status? Explain.

**1 mark**

**Total 10 marks**

**Question 4**

The owner of the health club wants to introduce a new weight-loss program and looks for a relationship between weight and fat percentage of participants.

**a** Identify the explanatory and response variables.

**1 mark**

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**b** Draw a scatter plot showing the relationship between these two variables.

**3 marks**

**c** Describe the scatter plot in terms of form, direction and strength. Does there appear to be a relationship between these two variables? Calculate Pearson's correlation coefficient to confirm.

**2 marks**

Now, consider the points on the scatter plot that belong to the women.

**d** Redraw the scatter plot, using only the female data.

**3 marks**

**e** Describe the relationship that appears between these points in terms of form, direction and strength.

**1 mark**

**f** Find Pearson's correlation coefficient, correct to four decimal places.

**1 mark**

**g** Find the equation of the least squares regression line and mark it on your scatter plot.

**2 marks**

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**h** Interpret the coefficient of determination.

**1 mark**

**i** Predict the weight of a woman (correct to one decimal place) with a body fat percentage of:

**i** 25.8%

**ii** 13.5%.

**2 marks**

**j** Which prediction do you have most confidence in? Explain.

**1 mark**

**k** Construct a residual plot and use this to determine whether the relationship is linear.

**4 marks**

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**l** Explain why, in order to see a pattern, each gender's points need to be considered separately.

**1 mark**

**Total 22 marks**

**Question 5**

Enrolments at the health club over 5 years are recorded below.

| Year | Jan | Apr | Jul | Oct |
|---|---|---|---|---|
| 1 | 52 | 43 | 32 | 46 |
| 2 | 67 | 51 | 39 | 56 |
| 3 | 92 | 55 | 46 | 68 |
| 4 | 88 | 71 | 49 | 65 |
| 5 | 105 | 77 | 68 | 91 |

**a** Draw a time series plot using the table of data above.

**3 marks**

**b** Describe the trend shown in the plot.

**1 mark**

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**c** Perform a three-point moving average to smooth the data and plot the resulting data on the same graph as the original data.

**3 marks**

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**d** Perform centred four-point moving median smoothing to smooth the data and plot the resulting data on the same graph as the original data.

**3 marks**

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**e** Which smoothing method provides the clearer view of the trend of the data? Explain.

**2 marks**

**f** Calculate the seasonal index (SI) for each quarter and deseasonalise the data.

**4 marks**

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**g** Plot the deseasonalised data along with the original data on a single graph.

**2 marks**

**h** Find the least squares regression line for the deseasonalised data.

**1 mark**

**i** Use this regression line to find the predicted actual enrolment numbers for the summer of 2020.

**2 marks**

**j** If the health club can cater for 150 people at a time only, will they be able to cater for the predicted number of enrolments in the summer of 2020? Explain.

**1 mark**

**Total 22 marks**

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